

Experiment 10: Version Control using Git & GitHub

Aim

To understand and perform version control using Git and GitHub.

Theory

Version control is a system that helps track changes in files over time. It allows multiple people to work on the same project without conflicts.

Git is a distributed version control system used to track changes locally.

GitHub is an online platform used to store and manage Git repositories.

Step-by-Step Procedure

1. Install Git and verify using: `git --version`
2. Configure username and email using `git config`
3. Create local repository using `git init`
4. Create files (e.g., `file1.txt`)
5. Add files using `git add`
6. Commit changes using `git commit`
7. Create a repository on GitHub
8. Link local repo to GitHub using `git remote add`
9. Push files using `git push`

Screenshots (Paste Below Each)

1. Git installation verification (git version)

```
Command Prompt
Microsoft Windows [Version 10.0.26100.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rushi>git --version
git version 2.54.0.windows.1

C:\Users\rushi>git config --global user.name "Rushikesh"

C:\Users\rushi>git config --global user.email "rushikeshp7972@gmail.com"

C:\Users\rushi>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Rushikesh
user.email=rushikeshp7972@gmail.com

C:\Users\rushi>
```

2. Repository creation

```
C:\Users\rushi>mkdir git-practical

C:\Users\rushi>cd git-practical

C:\Users\rushi\git-practical>git init
Initialized empty Git repository in C:/Users/rushi/git-practical/.git/

C:\Users\rushi\git-practical>
```

3. Commit history

```
C:\Users\rushi\git-practical>git commit -m "First commit"
[master (root-commit) 6a6800d] First commit
 1 file changed, 1 insertion(+)
 create mode 100644 file1.txt

C:\Users\rushi\git-practical>git log
commit 6a6800d503ce25b0c059b27d8c3c725713f9c252 (HEAD -> master)
Author: Rushikesh <rushikeshp7972@gmail.com>
Date:   Mon May 4 22:36:42 2026 +0530

    First commit

C:\Users\rushi\git-practical>
```

4. GitHub repository page

The screenshot shows the GitHub interface for a repository named 'git-practical'. At the top, there are navigation buttons: Pin, Watch (0), Fork (0), and Star (0). Below this, the repository name 'git-practical' is displayed with a 'Public' badge. The main content area shows the 'main' branch with 1 branch and 0 tags. A search bar is present with the text 'Go to file'. Below the search bar, there is a commit history section showing a single commit by 'youknowme-xd' with the message 'First commit' and a timestamp of '7 minutes ago'. The commit details show a file named 'file1.txt' with the message 'First commit' and a timestamp of '7 minutes ago'. Below the commit history, there is a 'README' section with a large 'Add a README' button and the text 'Help people interested in this repository understand your project.' On the right side, there is a sidebar with sections: 'About' (No description, website, or topics provided), 'Activity' (0 stars, 0 watching, 0 forks), 'Releases' (No releases published, Create a new release), 'Packages' (No packages published, Publish your first package), and 'Contributors' (1 contributor, youknowme-xd Rushikesh Patil).

5. Successful push

```
C:\Users\rushi\git-practical>git remote add origin https://github.com/youknowme-xd/git-practical

C:\Users\rushi\git-practical>git remote -v
origin  https://github.com/youknowme-xd/git-practical (fetch)
origin  https://github.com/youknowme-xd/git-practical (push)

C:\Users\rushi\git-practical>git branch -M main

C:\Users\rushi\git-practical>git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 224 bytes | 224.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/youknowme-xd/git-practical
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\rushi\git-practical>
```

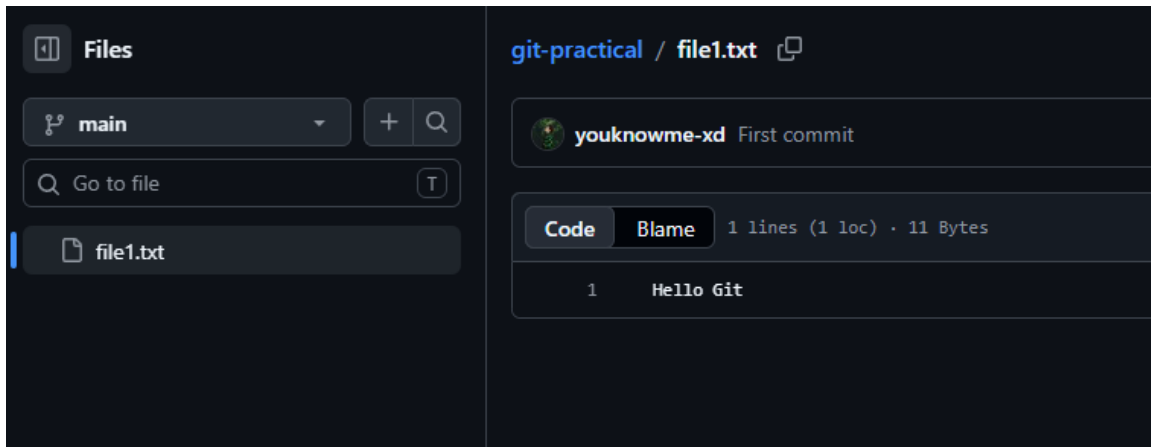
6. Git log output

```
C:\Users\rushi\git-practical>git log
commit 6a6800d503ce25b0c059b27d8c3c725713f9c252 (HEAD -> main, origin/main)
Author: Rushikesh <rushikeshp7972@gmail.com>
Date:   Mon May 4 22:36:42 2026 +0530

    First commit

C:\Users\rushi\git-practical>
```

7. Files visible on GitHub



Questions & Answers

1. Difference between Git and GitHub

Git is a version control tool used locally. GitHub is a platform to host repositories online.

2. Difference between git add and git commit

git add adds files to staging area. git commit saves changes permanently.

3. What is the staging area in Git?

It is a temporary area where files are prepared before committing.

4. What happens if you modify a file after committing it?

Git marks it as modified and you need to add and commit again.

5. Difference between git push and git pull

git push uploads changes to GitHub. git pull downloads changes from GitHub.

6. What is Version Control? Centralized vs Distributed VCS

Version control tracks file changes.

Centralized: one server (e.g., SVN).

Distributed: full copy with each user (e.g., Git).